



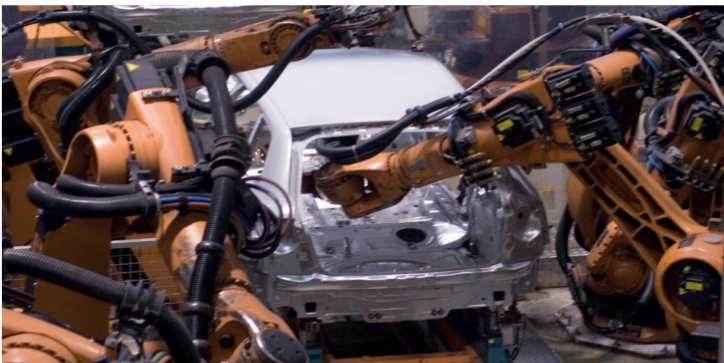
A-Trode®

The universal cap electrode for
resistance welding of light-to-medium
gauge coated and uncoated steels

A-Trode® production

Produced by Luvata's proprietary billet casting process, A-Trode's freedom from oxygen allows us to alloy the copper with the optimum levels of chrome and zirconium. Electrodes are cold headed or machined to suit the required shape.

Luvata has a long history of supplying high quality CuCrZr (A-Trode) electrodes.

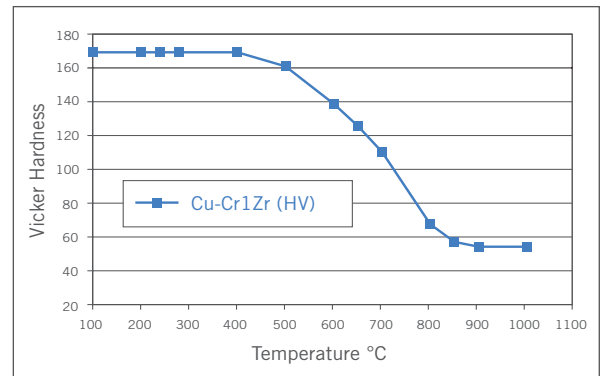


About Luvata

Luvata is a world leader in copper manufacturing and related engineering services. It brings together people, innovation and technology to create solutions in areas such as e-mobility, healthcare, and in electrification & the green transition. The company's success is driven by innovation, lean production and value-added sales, and its vision is to provide copper solutions which benefit people, society and the earth. Employing over 1,400 staff on four continents, Luvata works in partnership with customers such as ABB, CERN and Siemens. Luvata is a group company of Mitsubishi Materials Corporation.

Quality Properties

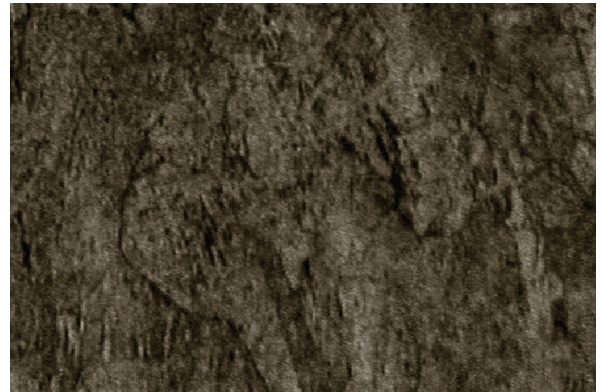
Alloy	C18150 CuCrZr, EN ISO 5182 A2/2, DIN 17666 Wn 2.1293, RWMA Class II
Physical material properties at 20°C	Mass Specific heat Thermal conductivity Expansion coefficient (20-300°C) Softening temperature 8,9g/cm³ 376 J/kg.K 320 W/m.K 17,0 x 10⁻⁶ m/mK min. 500°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 possible if desired against a charge.
Area of application	Male and female resistance welding electrodes Backing dies - series and indirect Seam welding wheels Projection welding electrodes



A-Trode® metallurgy



Conventional CuCrZr grain structure



Luvata's CuCrZr fine grain structure

Mechanical Properties

Form of supply	Tensile strength [N/mm ²]	0.2% offset yield strength [N/mm ²]	Elongation AS [%]
Electrodes	≥ 490	≥ 430	≥ 15

Physical Properties

Hardness at ambient temperature	Minimum 75 HRB
Conductivity	Minimum 80% IACS

Errors and omissions excepted. Values given are published industry material standards.

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Kova®

The high-strength, beryllium-free alloy

Kova is a 'beryllium-free' alloy made of copper, nickel, silicon and chromium (CuNiSiCr). Kova is the perfect replacement for harmful beryllium copper alloys.

Kova is a very high-strength, beryllium-free precipitation hardening alloy with medium-high electrical and thermal conductivity.

Advantages

- No harmful ingredients
- Easy disposal
- Easy to machine and form
- High strength with moderate thermal and electrical conductivity
- Extended use due to hardness

Applications

- Adaptors
- Shanks
- Projection welding electrodes



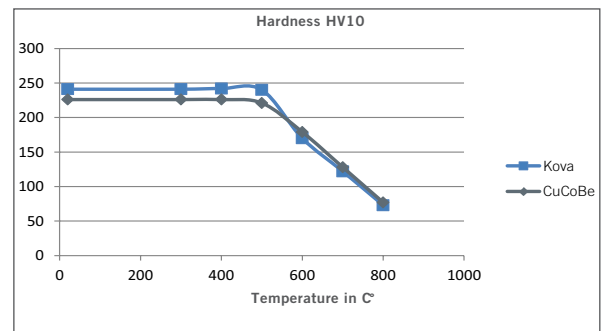
About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution.

The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 6 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

Specification

Alloy	Kova® NK203 is a RWMA class III alloy		
Chemical composition	Ni 1.8 - 3.0%; Si 0.4% - 0.8%; Cr 0.1 - 0.8% Fe 0.15% max; Cu balance		
Physical, electrical and thermal properties	Density	8.8kg/dm ³	
	Coefficient of linear expansion	0.0000175 1/K	
	Specific heat	380 J/(kg x K)	
	Melting temperature	1020-1040°C	
	Electric conductivity	22-24 MS/m	
	Electrical conductivity (% IACS)	38-42%	
	Electrical resistivity (mass)	0.37-0.41 g/m ³	
	Thermal conductivity (20°C)	220 W/Km	
Documentation	Acceptance test certificate N-CEN/TS 13388:2008		

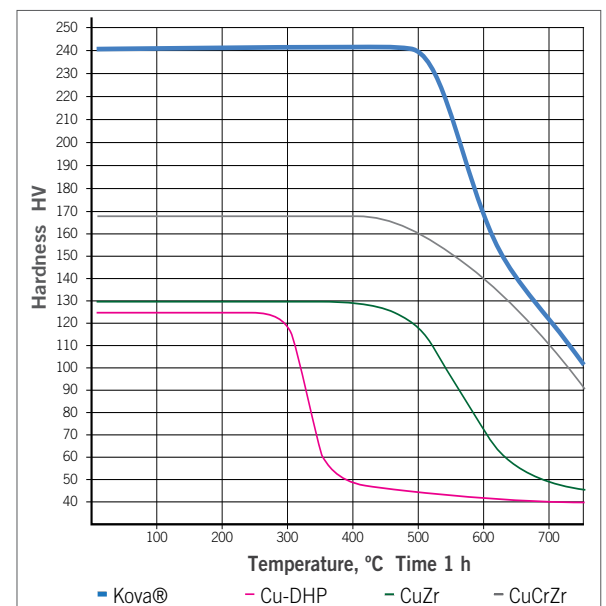
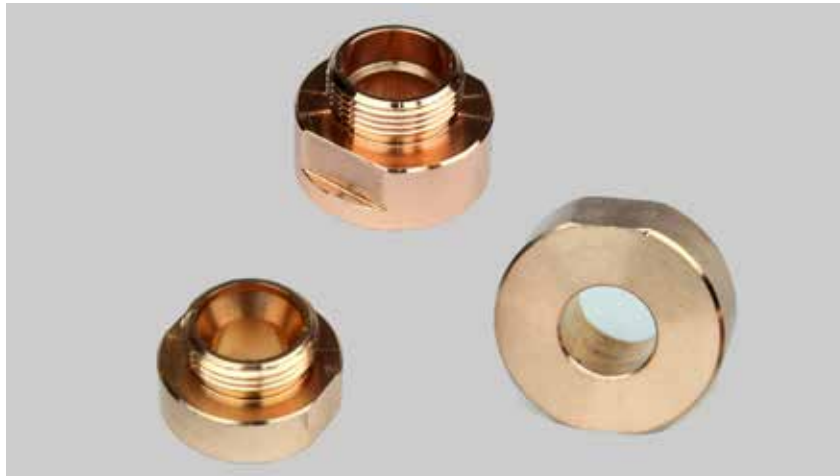


Kova hardness comparison

Mechanical Specifications

Form of supply	Tensile strength [N/mm ²]	0.2% offset yield strength [N/mm ²]	Elongation AS [%]	Hardness HV
Adaptors	650-800	600-750	9-15	220-250

Errors and omissions excepted. Values given are industry standards.



Kova softening behavior - resistance against softening

Room temperature hardness is presented in the figure above as a function of annealing temperature. Material at hard or aged temper.

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Nitrode®

No other cap electrode can match Nitrode® for its cost-effective, high quality performance in resistance-welding applications

The Nitrode cap is a cold-formed alloy of copper dispersion strengthened with aluminum oxide. It consistently outperforms copper chrome and copper chrome zirconium electrodes in resistance to annealing, consistent electrical conductivity, electrode life and lower maintenance costs.

Longer weld life

Nitrode caps last longer than conventional electrodes by resisting annealing. The contact surface of a resistance welding electrode reaches temperatures of up to 900°C during the welding process. Nitrode electrodes resist the mushrooming effect on the contact surface of the electrode, allowing more welds before tip dressing, heat stepping or tip change.

Non-stick

Nitrode caps reduce sticking of electrodes on galvanized steel and other coated metals. The refractory qualities of Al_2O_3 reduce the infusion of liquid and gaseous zinc into the copper matrix.

Reduces energy requirements

Nitrode requires lower current when used on both sides of the weldment. Current settings on your welder can be reduced by up to 10% from conventional settings, with no loss of weld quality. Nitrode allows for smoother startups with no warm ups, meaning fewer electrode changes and fewer interruptions for less maintenance downtime than other conventional electrodes.

Works on all steels

Nitrode has demonstrated superior welding performance for a variety of steels, including HSS, HSLA, micro-alloyed, nitrogenized, low-carbon, electrolytically zinc-coated, galvanized and others.

Nitrode® cuts your company's costs

Nitrode resists mushrooming for improved up-time from reduced tip changes, less tip-dressing, and lower current settings will increase your company's productivity, and cut costs.

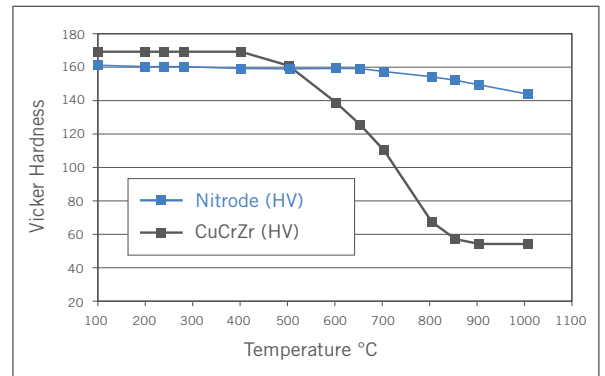


About Luvata

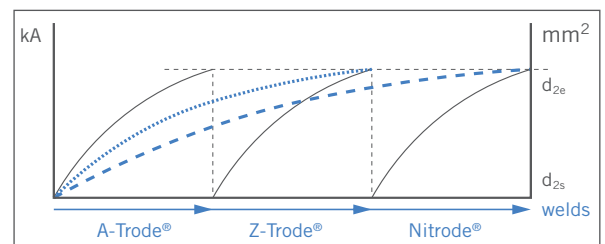
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Quality Properties*

Alloy	C15760 CuAl ₂ O ₃ , EN ISO 5182 C20/1, RWMA Class 20
Chemical composition	Aluminum 0.60% by weight as Al ₂ O ₃ Copper balance
Physical material properties at 20°C	Density 8.81g/cm ³ Thermal conductivity 322 W/m.K Expansion coefficient (20-150°C) 16.6 x 10 ⁻⁶ m/mK Electrical conductivity 78% IACS Softening temperature 1083°C
Dimensions and tolerances	To ISO 5821 or other standards as required. Special electrodes to customer drawing.
Packaging	Most items in cartons of 500 pieces.
Documentation	Acceptance test certificate EN 10204 3.1 B possible if desired against a charge.
Area of application	Male and female resistance welding electrodes Backing dies - series and indirect Projection welding electrodes



Hardness at high temperature



Heat stepping

Mechanical Properties*

Form of supply	Tensile strength MPa	Yield strength MPa	Elongation [%]	Hardness HV
Electrodes	489 - 537	393 - 441	18 - 23%	160

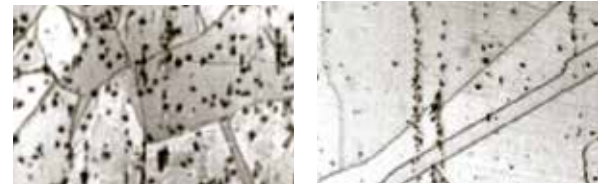
Physical Properties*

Hardness at ambient temperature	83 HRB
Electrical conductivity	78% IACS

*Errors and omissions excepted. Values given are industry standards. Actual properties will vary dependent upon amount of cold work.

Traceability

All materials are fully traceable. Nitrode electrodes can be recognized by their single knurls around the periphery of the electrode.



CuCrZr before and after annealing at 900°



Nitrode before and after annealing at 900°



CuCrZr after 1200 welds

Nitrode after 1200 welds

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